

Design of QR Based Smart Student Attendance System

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Abstract—Student attendance system is used to measure student participation in a classroom. Before pandemic attendance was taken manually like in sheets or registers. But when the pandemic hit, everything was online, so even the classes. The attendance count is a very important problem that the administrator needs to be more careful about taking during the online classes as there are many chances of a proxy happening. So, we came up with this proposed system “Student attendance using QR code” This paper proposes an attendance system that is based on the QR code-based attendance system. The students need to scan the QR in the class according to the professor instruction. By implementing this proposed system, we can reduce proxy and time in taking attendance of students. In order to design this proposed system, we are using technologies like OpenCV through python, and some libraries like MYQR, PYbase64, Pyzbar.

Keywords—QR code, Attendance system, Open CV, Smart Attendance

I. INTRODUCTION

When the manual attendance is taken by some university instructor during classes it's always a time-consuming process. This kind of process loses the interest of students when they are being taught some interesting topics or the general classes being taught. In other words, the total number of hours consumed per subject is almost 45 hours, but almost 5 to 6 hours are consumed in attendance. So, this may be a loss for lecturers and students as they lose their time and even sometimes portions of syllabus may not get completed on time. This may be a huge loss for students, who fail to score in their both end semester and internal exams. So, our proposed system gives every student a unique QR code containing the credentials of the students and it should be scanned by the scanner and it automatically marks attendance for students. QR (Quick response) was first designed in the automotive industry in Japan. Nowadays the QR code system has become popular these days due to it taking less time and greater storage than other technology like Universal Product Code (UPC) barcodes. A QR code is a machine readable that

stores the info about the documents. The code consists of Black square dots arranged in a box-like grid that looks like a square shape on a white background.



Fig-1: Sample QR Code of Student 1

A QR is read by some scanning devices like cameras and gets formatted algorithmically by underlying software using REED-Solomon error correction until the image can be scanned and data gets extracted both the ways horizontally and vertically of the code. We have created a unique QR code for every student. First, we are creating a text file. That text file consists of name of every student in the class. Then, we import pybase, pyzbar and myqr module inside python. After importing these modules, we are generating QR for every students that saved in the same project file. Three samples of QR code generated by us are shown in Fig 1-3. We are using video capture function of OpenCV to scan the QR. Initially, we need to import openCV. We may use OpenCV for any type of image processing activities. Here, scanning of QR for taking attendance will done by professor. We are using

imshow to show the image and we fix a time interval, so our proposed system will wait for QR code for those fixed duration.



Fig-2: QR Code of Student 2



Fig-3: QR Code of Student 3

II. LITERATURE REVIEW

In [1], they have been tried with students having to scan the QR code in a classroom. As technology is improving students have a chance to login with friend's details at the same time trying for a proxy. So, in our proposed system students have a different unique QR code which can't be shared to other devices they have to show the QR to the webcam on online classes. This process can reduce the proxy attendance. In [2], their proposed system used figure print technique for the attendance in a class. A biometric machine scans a fingerprint for the verification of a student that takes a long process if there are more than 60 students in a class. This process consumes a lot of time when an interesting lecture is going on. In [3], researchers are using real time computer vision algorithms in automated attendance management systems. They also used automated attendance management that has real time face detection using algorithms. Proposed system in

[3] automatically detects the registered students and marks their attendance. And what if the student leaves after marking the attendance? This proposed system doesn't have the details like how many minutes they have logged in to the class. So, from referring to all these research papers we have proposed a system that gives benefits to the students as well as teachers or mentors. A QR based system is developed by an automation industry in Japan which is trending now for payments across the world. QR is a user-friendly software. This proposed project takes the student attendance using QR. Each student is provided with a unique code that contains the details of the students which is already stored in the database. Students just have to show the code in front of a scanner or a cam which notes the attendance as per dates and time. Initially, we open the website which is shared through the group of the class like WhatsApp. If the website is opened for the first time, we need to register on the website by filling in the correct details like email or number and username along with password. Secondly, we need to enter the login details to enter the related details like name, roll number, mobile number, upload photo, email, class, branch etc. and proceed for the further step to get a python generated QR code which is unique for every individual student. In [4], they described how they used QR-Codes in place of online card swapping to help the tracking of class attendance. In [5], describe the ongoing attendance system of University of Sulaimaniyah, that used a QR code scanned by smart phone. Also, student will able to see their attendance record for every course they are taking in their running semester. Their inhouse system is so portable, that can be easily use by any other school of university. In [6], they are not only using QR system for attendance management but also for evaluation of teaching pedagogy. In [7], researcher developed an in-house learning management system that integrates features of attendance management using QR codes and institutional information system. In [8], there is a case study of using QR Codes in Romblon State University-Cajidiocan Campus for taking the attendance by professor or instructor.

III. METHODOLOGY

Nowadays, almost all university using QR Code for attendance. They are usually integrating this QR based attendance system inside their learning management system. The QR code is becoming very famous in this decade. The main reason behind this popularity is improvement in optical power of Mobile Phones. Every mobile phone is now equipped with QR-code reader. In our proposed methodology, we are generating QR code for every student mention in text file called student.txt and then share those generated QR to students. Finally, we scan the QR. We mark the attendance based on matching of QR. Finally, we store attendance in database (here, in our project, that is a text file attendance.txt) as shown in Fig. 4. In QR generation, L (7%), M(15%), Q(25%) and H (30%) are four levels of error corrections. We are using level H for generation for our QR code. There are forty (40) different versions of the QR code regualte the size of QR Code. Version 1 is the smallest, whereas version 40 is the largest. We are using Version 1 in our python code and generating a 21x21 matrix QR code. We are using getcwd function to get current working directory, to save our generated QR in the same current working directory. Finally, we decode the QR code with the help of python library called pyzbar. Decoding the QR code means we are getting the name of students. Eventually, we are using destroyallwindows to delete all windows, we have created during execution of this project.

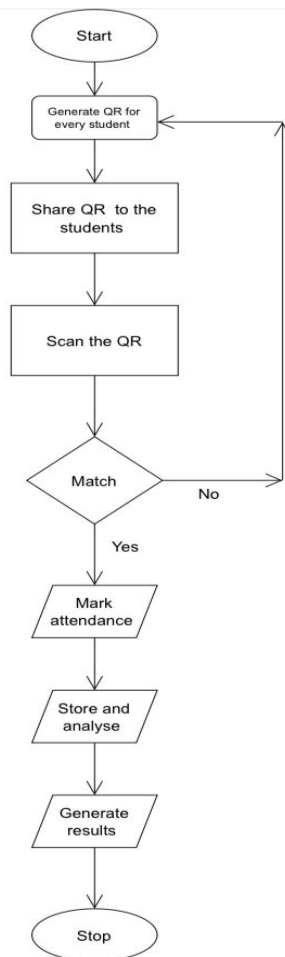


Fig-4: Flowchart OF QR Based Student Attendance System

into the database. Here the database cross checks both the stored and scanned QR's and marks the attendance. If the details are mismatched it shows that the details are incorrect, and if the registered students have not attended the class, they can't show the QR code, so after the completion of class it automatically makes them absent.

IV. RESULT

The developed system presented in this presentation has been successfully designed and tested. The student's attendance status was analysed thoroughly. Attendance monitoring systems are important for keeping track of attendance during lectures. Attendance recorded using QR codes are more accurate in comparison to the manual method of role calls. In this PCL report we have given an introduction of the Attendance monitoring system and its advantages. This system has proved to be an efficient method to record student attendance by students using their smartphones via web and thus helping save time and effort of the teacher role calling students individually by their name. The initial stage of our proposed system is shown in Fig. 6.

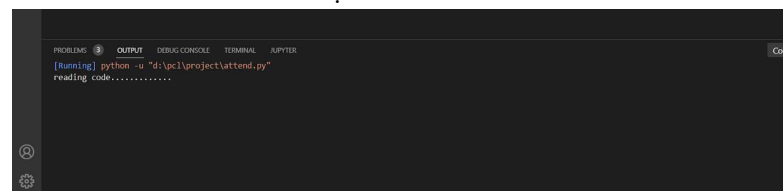


Fig-6: The Initial Stage of Our Proposed System

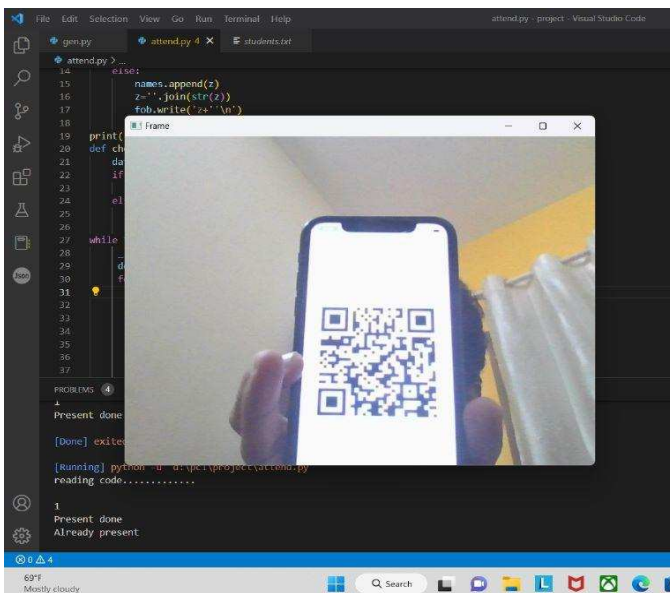


Fig-5: QR Scanner Interface

QR Scanner Interface is shown in Figure 5. The provided details will be stored in the database and cannot be changed without the permission of the administrator. Every student gets his own QR to scan in front of the scanner interface in order to record the attendance. After every lecture the teacher takes the attendance via scanner and uploads the scanned QR

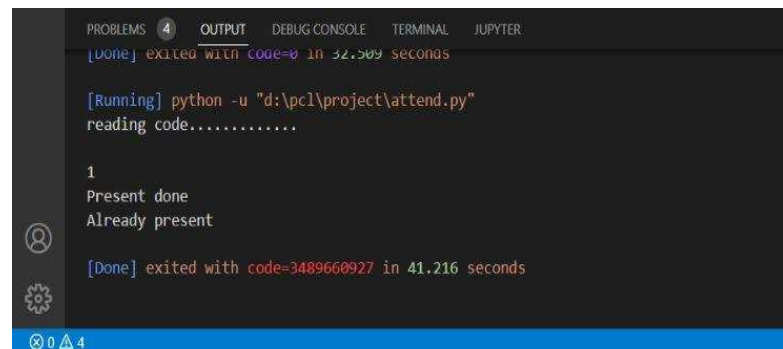


Fig-7: Final Output of Our Proposed System

The Final Output of our proposed system is shown in Fig. 7. The above picture shows that the attendance of the student has been recorded and stored and generated.

V. CONCLUSION

These days it is necessary to keep up with the latest technologies, especially in schools and colleges. Educational intuitions have been looking for these latest technologies for attendance. Educational institutions have been looking for ways to enhance the educational process using the latest technologies. Looking at the present situation we have come up with this methodology to improve in the field of

attendance. Due to this the accuracy of the attendance system increases. It mainly benefits in saving time and helps to improve the lecture environment. In the other words the lecture doesn't need to take extra classes to complete their portion. The system is able to manage student attendance records efficiently and there will be no use of carrying a physical attendance register; the proposed system detects fraud and improves accuracy and time.

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